

1 - Assignment: Booting 1

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▼ Task 1 - PXE Installation

PXE Server Setup

1. I created the first virtual machine using VirtualBox (**PXESERVER**) and isolated a private network on my workstation.

Below on figure 1 configuration for image, name, memory, credentials.

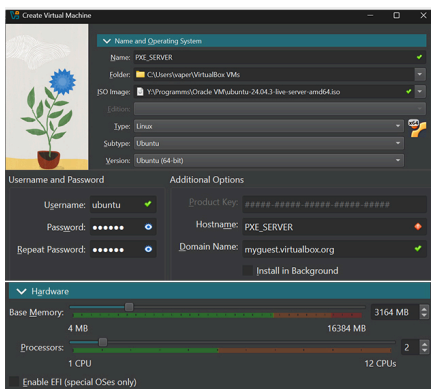


Figure 1. VM configuration

I used two network adapters, figure 2.

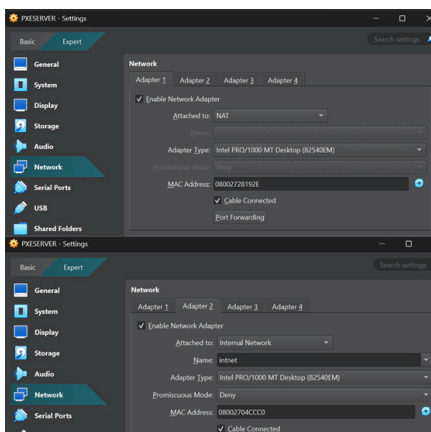


Figure 2. Network configuration

```
ubuntu@PXESERVER:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:28:19:2e brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86004sec preferred_lft 86004sec
    inet6 fd17:625c:f037:2:a00:27ff:fe28:192e/64 scope global dynamic mngtppaddr noprefixroute
        valid_lft 86390sec preferred_lft 14390sec
    inet6 fe80::a00:27ff:fe28:192e/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 08:00:27:04:cc:c0 brd ff:ff:ff:ff:ff:ff
```

Figure 2. Network configuration (ip addresses of adapters)

```
GNU nano 7.2 /etc/netplan/00-installer-config.yaml
network:
  ethernets:
    enp0s3:
      dhcp4: true
    enp0s8:
      dhcp: false
      addresses: [192.168.1.1/24]
  version: 2
```

Figure 2. Network configuration (`netplan` configuration for DHCP (please note this version is with syntax errors, which I fixed later))

```
ubuntu@PXESERVER:~$ sudo netplan apply
** (generate:2213): WARNING **: 11:42:08.428:
by others.
```

Figure 2. Network configuration (applying netplan configuration)

Installing tools and packages

2. To set up some of the services, such as DHCP, TFTP, HTTP, NFS I used `dnsmasq.service`.

```
sudo apt install dnsmasq-base
```

```
sudo apt install dnsmasq
```

```
sudo apt install apache2 net-tools unzip -y
```

```
ubuntu@PXESERVER:~$ sudo apt install dnsmasq-base
(sudo) password for ubuntu:
```

Figure. Installing dnsmasq

I install **Ubuntu 22.04.5 live-server** ISO on the PXE server inside **/tmp** folder.

```
ubuntu@PXESERVER:~$ wget https://releases.ubuntu.com/22.04/ubuntu-22.04.5-live-server-amd64.iso
--2025-12-09 11:43:53-- https://releases.ubuntu.com/22.04/ubuntu-22.04.5-live-server-amd64.iso
Resolving releases.ubuntu.com (releases.ubuntu.com)... 91.189.91.108, 91.189.91.107, 91.189.91.109,
Connecting to releases.ubuntu.com (releases.ubuntu.com)|91.189.91.108|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2136926208 (2.0G) [application/x-iso9660-image]
Saving to: 'ubuntu-22.04.5-live-server-amd64.iso'

ubuntu-22.04.5-live-server-amd64.iso      2%[=>]
--2025-12-09 11:44:04 (5.60 MB/s) - Read error at byte 60911877/2136926208 (Connection reset by peer)
--2025-12-09 11:44:05-- (try: 2) https://releases.ubuntu.com/22.04/ubuntu-22.04.5-live-server-amd64.iso
Connecting to releases.ubuntu.com (releases.ubuntu.com)|91.189.91.108|:443... connected.
HTTP request sent, awaiting response... 206 Partial Content
Length: 2136926208 (2.0G), 2076014331 (1.9G) remaining [application/x-iso9660-image]
Saving to: 'ubuntu-22.04.5-live-server-amd64.iso'

ubuntu-22.04.5-live-server-amd64.iso      2%[==>]
ubuntu-22.04.5-live-server-amd64.iso      5%[==>]
```

Figure.

Mount ISO and extract boot files to **/srv/tftp/ubuntu/**.

```
--2025-12-09 11:49:41 (5.90 MB/s) - 'ubuntu-22.04.5-live-server-amd64.iso' saved [2136926208/2136926208]

ubuntu@PXESERVER:~$ sudo mkdir -p /mnt/iso /srv/tftp/ubuntu
ubuntu@PXESERVER:~$ sudo mount ubuntu-22.04.5-live-server-amd64.iso /mnt/iso
mount: /mnt/iso: WARNING: source write-protected, mounted read-only.
ubuntu@PXESERVER:~$ sudo cp /mnt/iso/casper/vmlinuz /srv/tftp/ubuntu/
ubuntu@PXESERVER:~$ sudo cp /mnt/iso/casper/initrd /srv/tftp/ubuntu/
ubuntu@PXESERVER:~$ sudo umount /mnt/iso
sudo: umount: command not found
ubuntu@PXESERVER:~$ sudo umount /mnt/iso
```

Figure. CLI for mounting ISO

Then I placed ISO in Apache directory for HTTP serving.

```
Index.html
ubuntu@PXESERVER:/var/www/html$ sudo mkdir -p /var/www/html/isos
```

Figure. Apache dir **isos** is created

```
ubuntu@PXESERVER:~$ cd ~
ubuntu@PXESERVER:~$ ls
ubuntu-22.04.5-live-server-amd64.iso
ubuntu@PXESERVER:~$ sudo cp ubuntu-22.04.5-live-server-amd64.iso /var/www/html/isos/
```

Figure. Copying iso to **isos** directory.

```
ubuntu@PXESERVER:/var/www/html/isos$ du ubuntu-22.04.5-live-server-amd64.iso
2086848 ubuntu-22.04.5-live-server-amd64.iso
ubuntu@PXESERVER:/var/www/html/isos$ sudo chown -R www-data:www-data /var/www/html/isos/
```

Figure. Permissions to `isos` directory.

Now I install a boot loader GRUB EFI and PXELINUX.

```
sudo apt install syslinux-efi grub-efi-amd64-bin -y
sudo apt install grub-efi-amd64-signed grub-efi-amd64-bin -y
sudo apt install pxelinux
```

Figure. CLI for copying bootloaders

```
ubuntu@PXESERVER:~$ ls /usr/lib/PXELINUX/
ipxeLinux.0 pxelinux.0
ubuntu@PXESERVER:~$ sudo cp /usr/lib/PXELINUX/pxelinux.0 /srv/tftp/

root@PXESERVER:/srv/tftp# cp /usr/lib/syslinux/modules/bios/ldlinux.c32 /srv/tftp/

ubuntu@PXESERVER:~$ sudo cp /usr/lib/PXELINUX/pxelinux.0 /srv/tftp/
ubuntu@PXESERVER:~$ ls /srv/tftp/
efi64 ldlinux.e64 pxelinux.0 ubuntu
ubuntu@PXESERVER:~$ ls /srv/tftp/efi64/
EFI
ubuntu@PXESERVER:~$ ls /srv/tftp/efi64/EFI/
BOOT
ubuntu@PXESERVER:~$ ls /srv/tftp/efi64/EFI/BOOT/
grubx64.efi
```

Figure 7 `/srv/tftp` folder

I created GRUB config (`/srv/tftp/grub/grub.cfg`).

```
sudo mkdir -p /srv/tftp/grub
sudo nano /srv/tftp/grub/grub.cfg
```

```
root@PXESERVER:~/share# cat /srv/tftp/grub/grub.cfg
set default=0
set timeout=5
if loadfont /grub/unicode.pf2 ; then
    set gfxmode=auto
    insmod efi_gop
    terminal_output gfxterm
fi
menuentry "Ubuntu 22.04 live (HTTP ISO)" {
    linux /ubuntu/vmlinuz ip=dhcp url=http://192.168.1.1/isos/ubuntu-22.04.5-live-server-amd64.iso --
    initrd /ubuntu/initrd
}root@PXESERVER:~/share#
```

Figure 9 `grub` setup

Create PXELINUX config for BIOS understanding (`/srv/tftp/pxelinux.cfg/default`).

```
root@PXESERVER:~/share# cat /srv/tftp/pxelinux.cfg/default
default ubuntu-live
label ubuntu-live
kernel /ubuntu/vmlinuz
append initrd=/ubuntu/initrd ip=dhcp url=http://192.168.1.1/isos/ubuntu-22.04.5-live-server-amd64.iso --root@PXESERVER:~/share#
```

Figure. `PXELINUX` config

Established a folder for TFTP containing PXE netboot files.

Detailed config for PXE netboot with the link to ubuntu iso which is served on apache2 web server. Ubuntu 22.04.05 server has been chosen as OS image. After getting the initial boot files, PXE client will download image from Apache2 http server and start the installation process

dnsmasq service

Dnsmasq configuration is displayed below, operating on the internal adapter enp0s8 in the isolated network. Settings for DHCP and TFTP are defined. UEFI features are incorporated too.

```
Dec 09 11:17:21 PXESERVER systemd-helper[1372]: dnsmasq: failed to create listen
Dec 09 11:17:21 PXESERVER dnsmasq[1372]: failed to create listening socket for >
Dec 09 11:17:21 PXESERVER systemd[1]: dnsmasq.service: Control process exited, >
Dec 09 11:17:21 PXESERVER dnsmasq[1372]: FAILED to start up
Dec 09 11:17:21 PXESERVER systemd[1]: dnsmasq.service: Failed with result 'exit>
Dec 09 11:17:21 PXESERVER systemd[1]: Failed to start dnsmasq.service - dnsmasq>
```

Figure. starting dnsmasq service (error)

Now I need to disable the service systemd-resolve that is on port 53 to start **dnsmasq**.

```
root@PXESERVER:~# sudo ss -lp "sport = :domain"
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port
Process
udp UNCONN 0 0 127.0.0.54:domain 0.0.0.0:*
users:(("systemd-resolve",pid=451,fd=16))
udp UNCONN 0 0 127.0.0.53%lo:domain 0.0.0.0:*
users:(("systemd-resolve",pid=451,fd=14))
tcp LISTEN 0 4096 127.0.0.54:domain 0.0.0.0:*
users:(("systemd-resolve",pid=451,fd=17))
tcp LISTEN 0 4096 127.0.0.53%lo:domain 0.0.0.0:*
users:(("systemd-resolve",pid=451,fd=15))
root@PXESERVER:~# systemctl stop systemd-resolved
root@PXESERVER:~# systemctl disable systemd-resolved
Removed "/etc/systemd/system/dbus-org.freedesktop.resolve1.service".
Removed "/etc/systemd/system/sysinit.target.wants/systemd-resolved.service".
^[[A^[[IDroot@PXESERVER:~# systemctl mask systemd-resolved
Created symlink /etc/systemd/system/systemd-resolved.service → /dev/null.
```

Figure. Resolving issue with dnsmasq



To undo my actions I would run `sudo systemctl defaults systemd-resolved`,

```
sudo systemctl unmask systemd-resolved
sudo systemctl enable systemd-resolved
sudo systemctl start systemd-resolved
```

I setup `dnsmasq` service and then configured it.

```

PXESERVER [Running] - Oracle VirtualBox
File Machine View Input Devices Help
GNU nano 7.2 /etc/dnsmasq.conf
# Fix CVE-2020-25684
dhcp-name-match=set:ignore,ignore-name,wpad

interface=enp0s8
bind-interfaces
domain=pxemaster.local

dhcp-range=enp0s8,192.168.1.10,192.168.1.192,255.255.255.0,1h
dhcp-option=option:router,192.168.1.1
dhcp-option=option:dns-server,192.168.1.1
dhcp-option=option:ntp-server,192.168.1.1
dhcp-match=set:efi64,option:client-arch,9
dhcp-match=set:efi64,option:client-arch,7
dhcp-boot=tag:efi64,efi64/EFI/BOOT/grubx64.efi
# For BIOS understanding
dhcp-boot=pxelinux.0,pxeserver,192.168.1.1

enable-tftp
tftp-root=/srv/tftp

pxe-service=X86PC, "PXELINUX (BIOS)", pxelinux
pxe-service=x86-64_EFI, "UEFI GRUB", efi64/EFI/BOOT/grubx64.efi

```

Figure. dnsmasq.conf

Then I restart: `sudo systemctl restart dnsmasq.service`

```

root@PXESERVER:~/share# sudo systemctl status dnsmasq.service
● dnsmasq.service - dnsmasq - A lightweight DHCP and caching DNS server
   Loaded: loaded (/usr/lib/systemd/system/dnsmasq.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-12-09 13:07:43 UTC; 6s ago
     Main PID: 1132 (dnsmasq)
       Tasks: 1 (limit: 4602)
      Memory: 3.2M (peak: 5.0M)
         CPU: 60ms
    CGroup: /system.slice/dnsmasq.service
            └─1132 /usr/sbin/dnsmasq -x /run/dnsmasq/dnsmasq.pid -u dnsmasq -r /run/dnsmasq/resolv.conf -7 /etc/dnsmasq.d/.dpkg-dist,.dpkg-old,.dpkg-new --local

Dec 09 13:07:43 PXESERVER systemd[1]: Starting dnsmasq.service - dnsmasq - A lightweight DHCP and caching DNS server...
Dec 09 13:07:43 PXESERVER dnsmasq[1132]: started, version 2.90 cachesize 150
Dec 09 13:07:43 PXESERVER dnsmasq[1132]: compile time options: IPv6 GNU-getopt DBus no-UBus i18n IDN2 DHCP DHCPv6 no-Lua TFTP contrack ipset nftset auth cryptd
Dec 09 13:07:43 PXESERVER dnsmasq-dhcp[1132]: DHCP, IP range 192.168.1.10 -- 192.168.1.192, lease time 1h
Dec 09 13:07:43 PXESERVER dnsmasq-dhcp[1132]: DHCP, sockets bound exclusively to interface enp0s8
Dec 09 13:07:43 PXESERVER dnsmasq-tftp[1132]: TFTP root is /srv/tftp
Dec 09 13:07:43 PXESERVER dnsmasq[1132]: read /etc/hosts - 0 names
Dec 09 13:07:43 PXESERVER resolvconf[1140]: Dropped protocol specifier '.dnsmasq' from 'lo.dnsmasq'. Using 'lo' (ifindex=1).
Dec 09 13:07:43 PXESERVER resolvconf[1140]: failed to set DNS configuration: Link lo is loopback device.
Dec 09 13:07:43 PXESERVER systemd[1]: Started dnsmasq.service - dnsmasq - A lightweight DHCP and caching DNS server.

root@PXESERVER:~/share# ping google.com
PING google.com (216.58.208.110) 56(84) bytes of data:
64 bytes from ans17s08-in-f14.1e100.net (216.58.208.110): icmp_seq=1 ttl=255 time=103 ms
64 bytes from ans17s08-in-f14.1e100.net (216.58.208.110): icmp_seq=2 ttl=255 time=102 ms

```

Figure. `dnsmasq.service`

```
root@PXESERVER:~/share# dnsmasq --test
dnsmasq: syntax check OK.
```

Figure. `dnsmasq.service` check

Start Apache.

```
root@PXESERVER:~/share# sudo systemctl enable --now apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/syst
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
root@PXESERVER:~/share# sudo systemctl status apache2
■ apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-12-09 13:00:26 UTC; 15min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 827 (apache2)
    Tasks: 55 (limit: 4602)
   Memory: 8.2M (peak: 8.6M)
      CPU: 157ms
   CGroup: /system.slice/apache2.service
           └─827 /usr/sbin/apache2 -k start
             829 /usr/sbin/apache2 -k start
             830 /usr/sbin/apache2 -k start

Dec 09 13:00:25 PXESERVER systemd[1]: Starting apache2.service - The Apache HTTP Server...
Dec 09 13:00:26 PXESERVER apachectl[812]: AH00558: apache2: Could not reliably determine t
Dec 09 13:00:26 PXESERVER systemd[1]: Started apache2.service - The Apache HTTP Server.
```

Figure. Apache started

```
root@PXESERVER:~/share# sudo netstat -tuln | grep 80
tcp6      0      0 fe80::a00:27ff:fe04::53 :::*           LISTEN
tcp6      0      0 :::80        :::*           LISTEN
udp6      0      0 fe80::a00:27ff:fe04::53 :::*
udp6      0      0 fe80::a00:27ff:fe04::69 :::*
root@PXESERVER:~/share# sudo ufw allow 80
Rules updated
Rules updated (v6)
```

Figure. Avoid any conflicts to access apache.

In-depth setup for PXE netboot including a reference to the Ubuntu ISO hosted on the apache2 web server. Ubuntu 22.04.5 server was selected as the OS image. Following the acquisition of the starting boot files, the PXE client will fetch the image from the Apache2 HTTP server and begin the setup procedure.

1.2.1. Write about each service's role in the PXE environment.

PXE server is a preboot execution environment ("pixie"), that allows booting OS on connected nodes (PCs) from the main Server directly - a different approach to the one manual, where we boot OS from the hard

disk of a machine.

PXE server is a client-server interface.

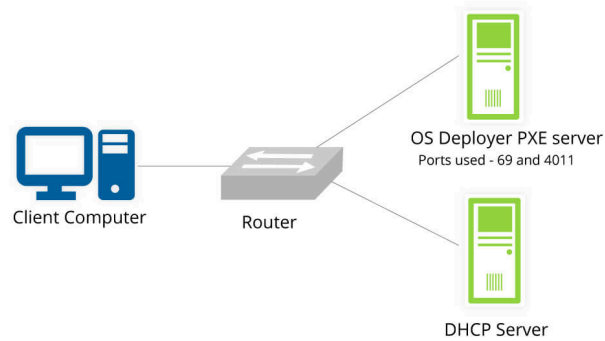


Figure. PXE architecture

1.3. Question: why not run your DHCP service on the SNE network directly?

Introducing a secondary DHCP server into an established network that already has a DHCP server could result in duplicate responses to DHCPDISCOVER queries from clients, and clients might experience disconnections due to ARP disputes.

PXE Client Setup

The client will automatically obtain an IP address from the DHCP server within the same isolated network as the PXE server. Afterward, the client retrieves netboot files from the PXE server through TFTP.

1. I created the second virtual machine using VirtualBox in order to test the PXE service.

It is important to note that I created empty VM: without an image, figure.

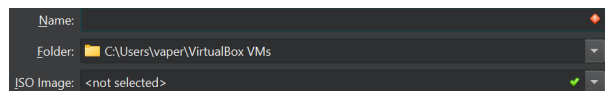


Figure. Config for Client VM

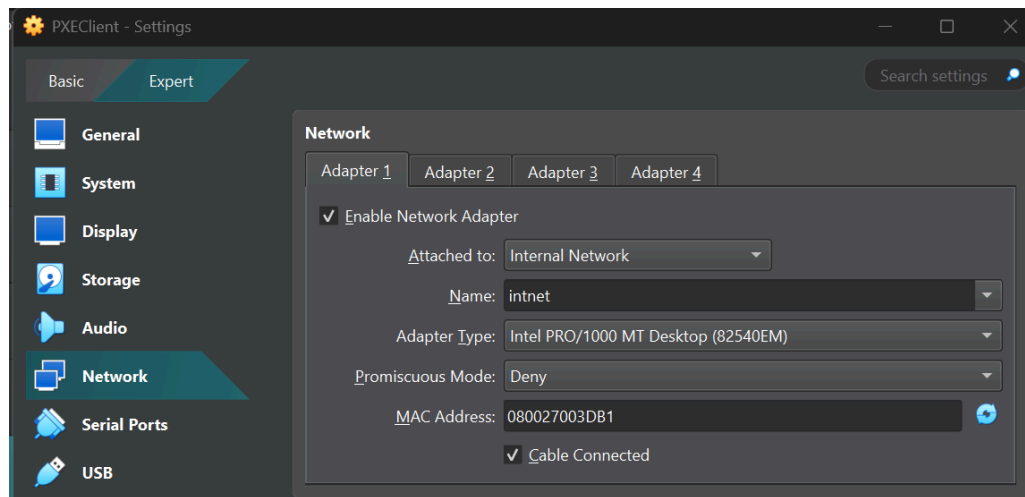


Figure. PXE client network adapter

1. Change the boot order

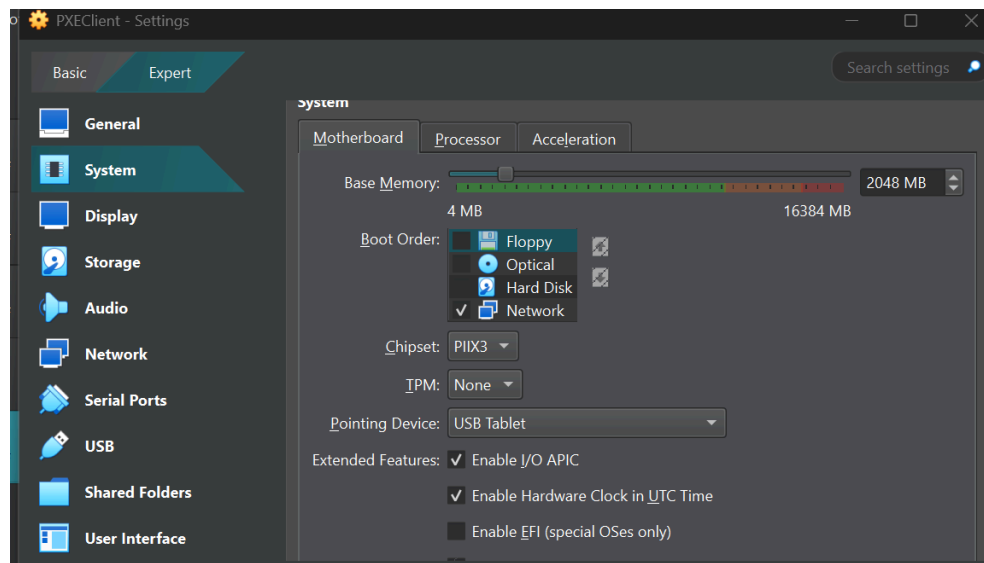


Figure. Chosen only network boot choice for PXE client

2. Show that your PXE client takes the IP, Figure PXE boot, client obtains IP from DHCP

```

: sent /srv/tftp/pxelinux.cfg/default to 192.168.1.183
: sent /srv/tftp/ubuntu/mlinuz to 192.168.1.183

CLIENT IP: 192.168.1.183

```

Figure. IP of client machine obtained from DHCP

3. Boot and install a new system with it and show the proof in the report.

```
Intel UNDI, PXE-2.1
PXE Software Copyright (C) 1997-2000 Intel Corporation
Copyright (C) 2010-2020 Oracle Corporation

CLIENT MAC ADDR: 08 00 27 00 3D B1 GUID: E5DA3853-4417-8848-B9AC-61D96D8BF801
CLIENT IP: 192.168.1.183 MASK: 255.255.255.0 DHCP IP: 192.168.1.1
GATEWAY IP: 192.168.1.1

Auto-select:
  PXELINUX (BIOS)

BOOT SERVER IP: 192.168.1.1

PXELINUX 6.04 PXE 20240408 Copyright (C) 1994-2015 H. Peter Anvin et al
Loading /ubuntu/vmlinuz... ok
Loading /ubuntu/initrd... _
```

```
PXEClient [Running] - Oracle VirtualBox
File Machine View Input Devices Help
[ 4.431502] vmwgfx 0000:00:02.0: [drm] DMA map mode: Caching DMA mappings.
[ 4.432077] vmwgfx 0000:00:02.0: [drm] Legacy memory limits: VRAM = 131072 kB
[ 4.432081] vmwgfx 0000:00:02.0: [drm] MOB limits: max mob size = 131072 kB,
[ 4.432085] vmwgfx 0000:00:02.0: [drm] Capabilities: rect copy, cursor, curso
[ 4.432096] vmwgfx 0000:00:02.0: [drm] Capabilities2: grow otable, intra surf
[ 4.432092] vmwgfx 0000:00:02.0: [drm] Max GMR ids is 8192
[ 4.432094] vmwgfx 0000:00:02.0: [drm] Max number of GMR pages is 1048576
[ 4.432096] vmwgfx 0000:00:02.0: [drm] Maximum display memory size is 131072
[ 4.434076] input: ImExPS/2 Generic Explorer Mouse as /devices/platform/i8042
[ 4.480297] vmwgfx 0000:00:02.0: [drm] Screen Target display unit initialized
[ 4.480685] vmwgfx 0000:00:02.0: [drm] Fifo max 0x00200000 min 0x00001000 cap
[ 4.481201] vmwgfx 0000:00:02.0: [drm] Using command buffers with DMA pool.
[ 4.481475] vmwgfx 0000:00:02.0: [drm] Available shader model: SM_5.
[ 4.481498] [drm:vmu_host_printf [vmwgfx]] *ERROR* Failed to send host log me
[ 4.482879] fbcon: svgadrmfb (fb0) is primary device
[ 4.484297] Console: switching to colour frame buffer device 100x37
[ 4.494829] [drm] Initialized vmwgfx 2.19.0 20210722 for 0000:00:02.0 on minor 0
[ 4.730564] e1000 0000:00:03.0 eth0: (PCI:33MHz:32-bit) 08:00:27:00:3d:b1
[ 4.730533] e1000 0000:00:03.0 eth0: Intel(R) PRO/1000 Network Connection
[ 4.732036] e1000 0000:00:03.0 enp0s3: renamed from eth0
Begin: Loading essential drivers ... [ 5.896155] clocksource: Long readout interval, skipping wat
chdog check: cs_nsec: 1182816320 wd_nsec: 1182815548
```

```
PXEClient [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Listening on LPF/enp0s3/08:00:27:00:3d:b1
Sending on LPF/enp0s3/08:00:27:00:3d:b1
Sending on Socket/fallback
DHCPDISCOVER on enp0s3 to 255.255.255.255 port 67 interval 3 (xid=0x73ff62)
DHCPOFFER of 192.168.1.183 from 192.168.1.1
DHCPREQUEST for 192.168.1.183 on enp0s3 to 255.255.255.255 port 67 (xid=0x62ff7300)
DHCPACK of 192.168.1.183 from 192.168.1.1 (xid=0x73ff62)
Bound to 192.168.1.183 -- renewal in 1790 seconds.
Begin: Running /scripts/casper-premount ... done.
done.
Begin: Trying netboot from 192.168.1.1: ... Begin: Trying to download and mount http://192.168.1.
1:192.168.1.1:80)
ubuntu-22.04.5-live- 5% [* | 110M 0:00:17 ETA
ubuntu-22.04.5-live- 13% **** | 266M 0:00:13 ETA
ubuntu-22.04.5-live- 20% ***** | 424M 0:00:11 ETA
ubuntu-22.04.5-live- 28% ***** | 571M 0:00:10 ETA
ubuntu-22.04.5-live- 31% ***** | 643M 0:00:10 ETA
ubuntu-22.04.5-live- 34% ***** | 707M 0:00:11 ETA
ubuntu-22.04.5-live- 37% ***** | 769M 0:00:11 ETA
ubuntu-22.04.5-live- 41% ***** | 835M 0:00:11 ETA
ubuntu-22.04.5-live- 43% ***** | 888M 0:00:11 ETA
ubuntu-22.04.5-live- 46% ***** | 944M 0:00:11 ETA
ubuntu-22.04.5-live- 49% ***** | 1010M 0:00:11 ETA
ubuntu-22.04.5-live- 52% ***** | 1079M 0:00:10 ETA
ubuntu-22.04.5-live- 56% ***** | 1151M 0:00:10 ETA
ubuntu-22.04.5-live- 60% ***** | 1223M 0:00:09 ETA

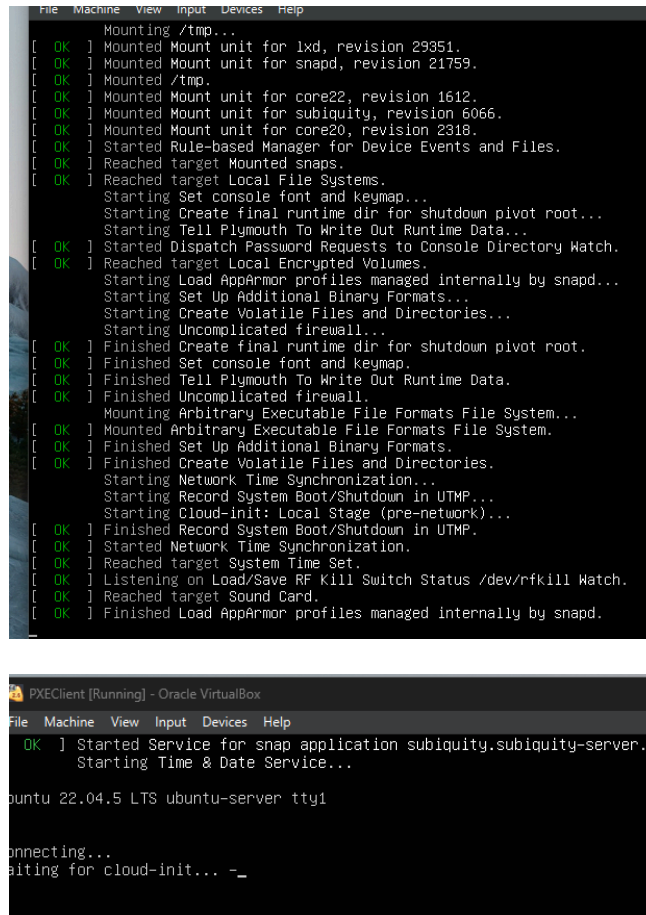
BusyBox v1.30.1 (Ubuntu 1:1.30.1-7ubuntu3.1) built-in shell (ash)
Enter 'help' for a list of built-in commands.
```

Figure. Booting and downloading image of OS on client machine

Upon choosing the installation choice, the image began downloading. As the disk remains unformatted, the image loads straight into RAM, requiring adequate RAM in the machine for a successful OS installation.

The Ubuntu 22.04.5 server image is approximately 2GB, so I allocated 8GB to the client machine as a precaution.

Following that, the installation commenced.



```
File Machine View Input Devices Help
[ OK ] Mounting /tmp...
[ OK ] Mounted Mount unit for lxd, revision 29351.
[ OK ] Mounted Mount unit for snapd, revision 21759.
[ OK ] Mounted /tmp.
[ OK ] Mounted Mount unit for core22, revision 1612.
[ OK ] Mounted Mount unit for subiquity, revision 6066.
[ OK ] Mounted Mount unit for core20, revision 2318.
[ OK ] Started Rule-based Manager for Device Events and Files.
[ OK ] Reached target Mounted snaps.
[ OK ] Reached target Local File Systems.
Starting Set console font and keymap...
Starting Create final runtime dir for shutdown pivot root...
Starting Tell Plymouth To Write Out Runtime Data...
[ OK ] Started Dispatch Password Requests to Console Directory Watch.
[ OK ] Reached target Local Encrypted Volumes.
Starting Load AppArmor profiles managed internally by snapd...
Starting Set Up Additional Binary Formats...
Starting Create Volatile Files and Directories...
Starting Uncomplicated firewall...
[ OK ] Finished Create final runtime dir for shutdown pivot root.
[ OK ] Finished Set console font and keymap.
[ OK ] Finished Tell Plymouth To Write Out Runtime Data.
[ OK ] Finished Uncomplicated firewall.
Mounting Arbitrary Executable File Formats File System...
[ OK ] Mounted Arbitrary Executable File Formats File System.
[ OK ] Finished Set Up Additional Binary Formats.
[ OK ] Finished Create Volatile Files and Directories.
Starting Network Time Synchronization...
Starting Record System Boot/Shutdown in UTMP...
Starting Cloud-init: Local Stage (pre-network)...
[ OK ] Finished Record System Boot/Shutdown in UTMP.
[ OK ] Started Network Time Synchronization.
[ OK ] Reached target System Time Set.
[ OK ] Listening on Load/Save RF Kill Switch Status /dev/rfkill Watch.
[ OK ] Reached target Sound Card.
[ OK ] Finished Load AppArmor profiles managed internally by snapd.

PXEClient [Running] - Oracle VirtualBox
File Machine View Input Devices Help
[ OK ] Started Service for snap application subiquity.subiquity-server.
Starting Time & Date Service...

ubuntu 22.04.5 LTS ubuntu-server tty1

Connecting...
Waiting for cloud-init... _
```

Figure. Early installation steps

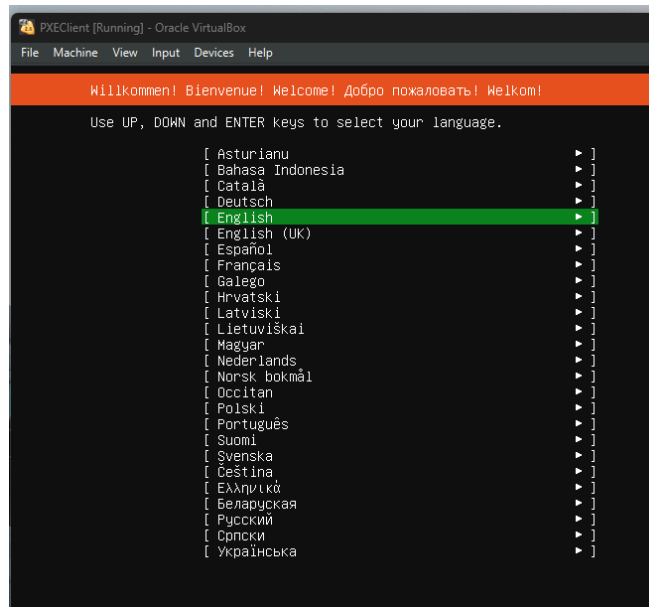


Figure. Language choice screen in installation

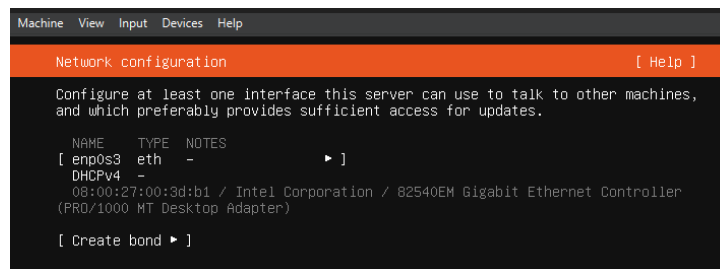


Figure. Network configuration.

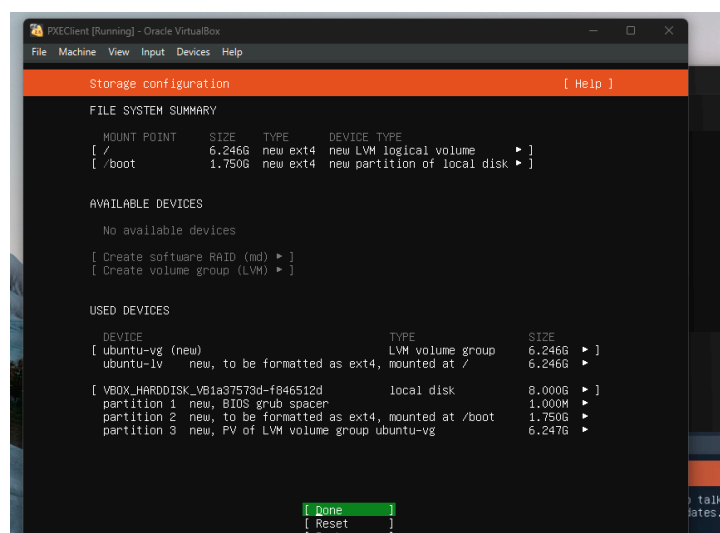


Figure. Disk setup screen

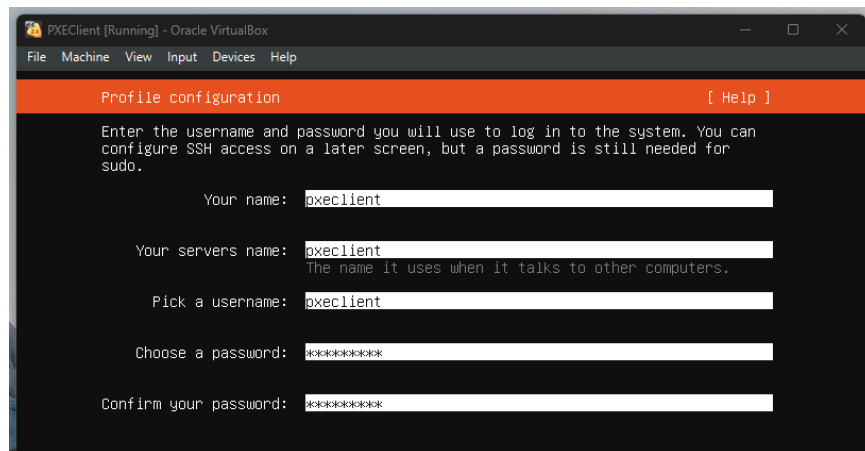


Figure. User profile setup

After some time, the installation finished without issues.

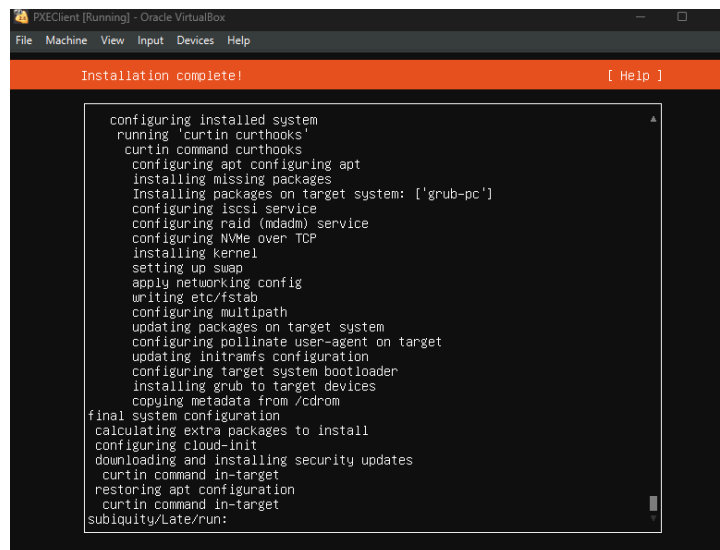


Figure. Installation success

Next, I restarted the machine from the hard drive to confirm the client's OS is functioning properly.

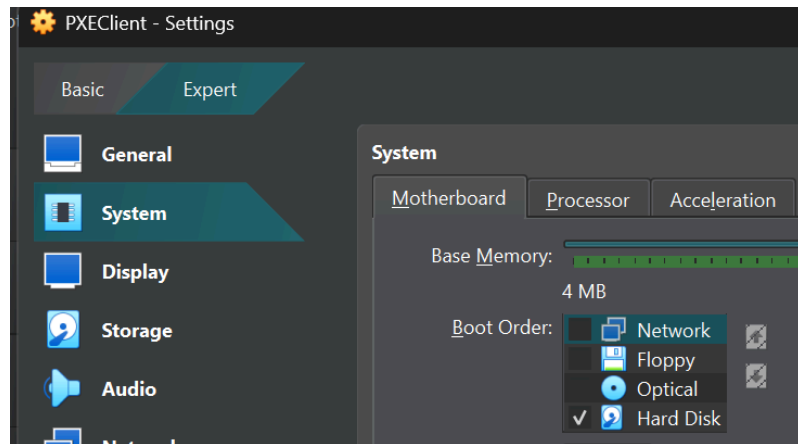


Figure. Started up the machine from hard disk

```

PXEClient [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Login incorrect
pxeclient login: pxeclient
Password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canoni
* Support:        https://ubuntu.com/pro

System information as of Tue Dec  9 04:43:

System load:  0.02          Processes
Usage of /:   48.2% of 6.06GB Users log
Memory usage: 2%           IPv4 addr
Swap usage:   0%

Expanded Security Maintenance for Applicati
0 updates can be applied immediately.

Enable ESM Apps to receive additional futur
See https://ubuntu.com/esm or run: sudo pro

The programs included with the Ubuntu syste
the exact distribution terms for each progr
individual files in /usr/share/doc/*/copyri

Ubuntu comes with ABSOLUTELY NO WARRANTY, t
applicable law.

To run a command as administrator (user "ro
See "man sudo_root" for details.

pxeclient@pxeclient:~$ _

```

Figure. Successfully logged in client machine.

```

root@PXESERVER:/srv/tftp# systemctl status dnsmasq.service
● dnsmasq.service - dnsmasq - A lightweight DHCP and caching DNS server
   Loaded: loaded (/usr/lib/systemd/system/dnsmasq.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-12-09 13:54:49 UTC; 41min ago
     Process: 1722 ExecStartPre=/usr/share/dnsmasq/systemd-helper checkconfig (code=exited, status=0/SUCCESS)
     Process: 1730 ExecStart=/usr/share/dnsmasq/systemd-helper exec (code=exited, status=0/SUCCESS)
     Process: 1736 ExecStartPost=/usr/share/dnsmasq/systemd-helper start-resolvconf (code=exited, status=0/SUCCESS)
   Main PID: 1735 (dnsmasq)
      Tasks: 1 (limit: 4602)
     Memory: 120.0M (peak: 120.5M)
        CPU: 1min 33.466s
   CGroup: /system.slice/dnsmasq.service
           └─1735 /usr/sbin/dnsmasq -x /run/dnsmasq/dnsmasq.pid -u dnsmasq -r /run/dnsmasq/resolv.conf -7 /etc/dnsm
Dec 09 14:26:28 PXESERVER dnsmasq-dhcp[1735]: DHCPDISCOVER(enp0s8) 08:00:27:00:3d:b1
Dec 09 14:26:28 PXESERVER dnsmasq-dhcp[1735]: DHCPPOFFER(enp0s8) 192.168.1.183 08:00:27:00:3d:b1
Dec 09 14:26:28 PXESERVER dnsmasq-dhcp[1735]: DHCPREQUEST(enp0s8) 192.168.1.183 08:00:27:00:3d:b1
Dec 09 14:26:28 PXESERVER dnsmasq-dhcp[1735]: DHCPACK(enp0s8) 192.168.1.183 08:00:27:00:3d:b1 ubuntu-server
Dec 09 14:33:49 PXESERVER dnsmasq-dhcp[1735]: DHCPDISCOVER(enp0s8) 08:00:27:00:3d:b1
Dec 09 14:33:49 PXESERVER dnsmasq-dhcp[1735]: DHCPPOFFER(enp0s8) 192.168.1.183 08:00:27:00:3d:b1
Dec 09 14:33:49 PXESERVER dnsmasq-dhcp[1735]: DHCPREQUEST(enp0s8) 192.168.1.183 08:00:27:00:3d:b1
Dec 09 14:33:49 PXESERVER dnsmasq-dhcp[1735]: DHCPACK(enp0s8) 192.168.1.183 08:00:27:00:3d:b1 pxecient
Dec 09 14:33:51 PXESERVER dnsmasq-dhcp[1735]: DHCPREQUEST(enp0s8) 192.168.1.183 08:00:27:00:3d:b1
Dec 09 14:33:51 PXESERVER dnsmasq-dhcp[1735]: DHCPACK(enp0s8) 192.168.1.183 08:00:27:00:3d:b1 pxecient

```

Figure. Status of `dnsmasq.service`

▼ Task 2 - Questions to answer

1. Briefly explain UEFI with secure boot enabled, UEFI without secure boot, and BIOS PXE booting approaches.
 - 1.1. How do they work? Explain with a simple diagram.

UEFI without secure boot offers identical UEFI functions and startup sequence, but skips signature validations, permitting unsigned loaders or modified kernels to proceed.

BIOS PXE relies on traditional firmware to obtain a network boot application and OS installer. It depends on DHCP/TFTP details transmitted over the connection, without inherent verification or protection for the boot application or following data, allowing unauthorized servers or intercepts to deliver malicious boot content.

UEFI Secure Boot

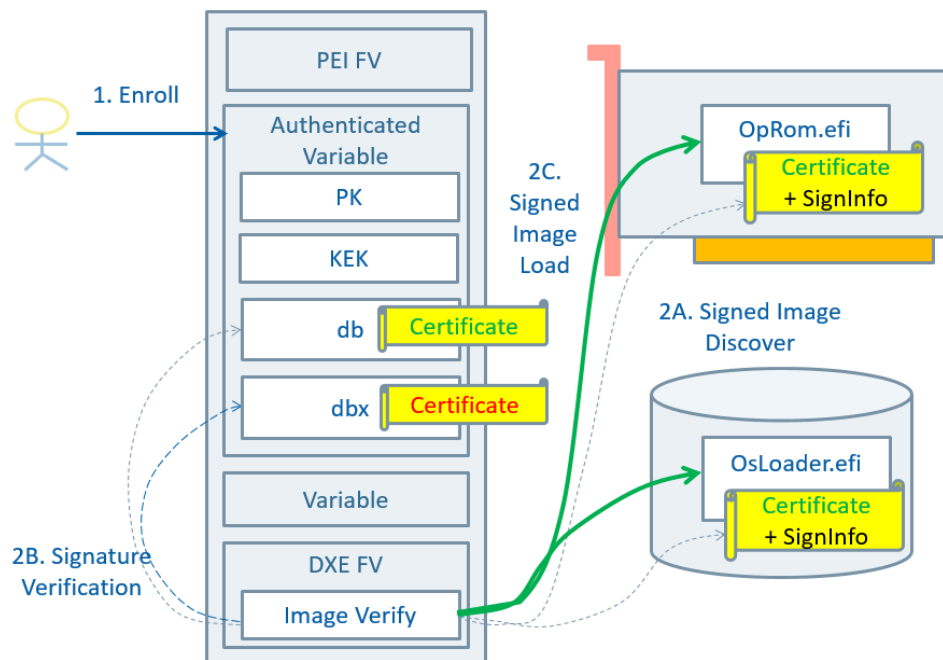


Figure. Secure boot flow

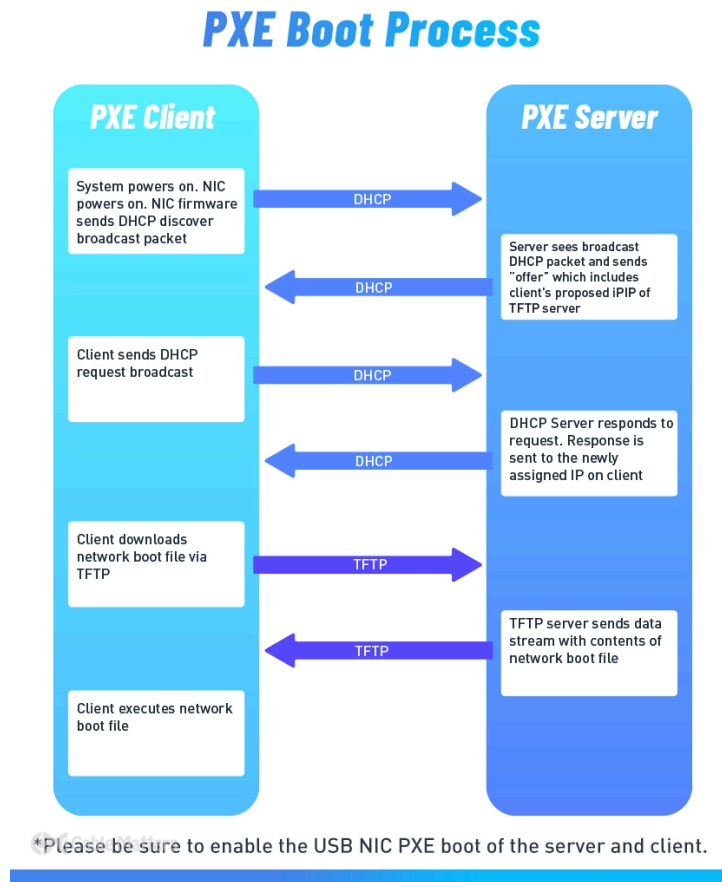


Figure. PXE boot flow

2. What is a GPT?

2.1. What is its general layout? Explain each element. What is the role of a partition table?

GPT represents the GUID Partition Table, a contemporary disk partitioning method that supersedes MBR, and forms part of UEFI.

The GPT employs a current Logical Block Addressing system. Inherited from its forerunner, LBA 0 holds the MBR, with the Primary GPT Header at LBA 1. After the GPT Header comes the partition table proper.

Overall structure:

- protective MBR (LBA 0): a basic MBR featuring one "protective" partition record spanning the disk to stop outdated MBR-only utilities from damaging GPT disks; it indicates GPT usage without actual partition info;
- primary GPT header (LBA 1): includes disk GUID, available LBAs, positions/sizes of the partition record array, and CRC32 checks for header and records to identify errors; this serves as the main guide at the

disk's beginning;

- partition record array (LBAs 2–33): a collection of uniform-size records outlining each partition: type GUID, distinct GUID, start and end LBAs, flags, and UTF-16 label;
- partitions (storage areas): the real data zones specified by the records, like an EFI System Partition (FAT32 for UEFI startup), OS, swap, backup, etc.;
- secondary partition record array and secondary GPT header (disk end): duplicate of the records and an alternate header at the final LBAs, allowing restoration if primary GPT data is corrupted.

The partition table acts as the definitive guide to the start and end points of each partition, its category, identifier, and properties, enabling firmware, loaders, and OS to find boot areas and filesystems while avoiding conflicts.

3. What is gdisk?

3.1. How does it work? What can you do with it? Provide a simple practice.

gdisk is a console-based partitioning utility from the GPT fdisk collection for generating, examining, and fixing GUID Partition Table (GPT) disks.

When launched, gdisk tries to detect the partitioning type on the disk. If valid GPT data is found, gdisk utilizes it.

It loads current disk metadata (protective MBR, primary GPT header, partition array, backup header), confirms CRC32 checks, and imports the table into memory for editable sessions that are only saved upon explicit approval.

It can convert legacy MBR partitions to GPT automatically.

gdisk primarily enables viewing, adding, removing, or adjusting GPT partitions, and modifying partition details.

Additionally, you can switch between formats, save and load GPT metadata, and conduct repair actions on tables.

```

C
root@PXESERVER:/srv/tftp# sudo gdisk -l /dev/sda
GPT fdisk (gdisk) version 1.0.10

Partition table scan:
  MBR: protective
  BSD: not present
  APM: not present
  GPT: present

Found valid GPT with protective MBR; using GPT.
Disk /dev/sda: 52428800 sectors, 25.0 GiB
Model: UBOX HARDDISK
Sector size (logical/physical): 512/512 bytes
Disk identifier (GUID): 05C14463-D21A-44D1-9349-41387FF5D505
Partition table holds up to 128 entries
Main partition table begins at sector 2 and ends at sector 33
First usable sector is 34, last usable sector is 52428766
Partitions will be aligned on 2048-sector boundaries
Total free space is 4029 sectors (2.0 MiB)

Number  Start (sector)    End (sector)  Size      Code  Name
   1            2048             4095     1024.0 KiB   EF02
   2            4096          52426751     25.0 GiB    8300
root@PXESERVER:/srv/tftp#

```

Figure. `gdisk` display

4. What is a Protective MBR and why is it in the GPT?

A Protective MBR is a simple Master Boot Record located at LBA 0 on a GPT disk that provides compatibility for older tools so they do not view the drive as blank or corrupt GPT data.

▼ Task 3 - Partitions

1. Verify the GPT schema of your Ubuntu machine.

```

root@PXESERVER:/srv/tftp# sudo gdisk /dev/sda
GPT fdisk (gdisk) version 1.0.10

Partition table scan:
  MBR: protective
  BSD: not present
  APM: not present
  GPT: present

Found valid GPT with protective MBR; using GPT.

Command (? for help): v

No problems found. 4029 free sectors (2.0 MiB) available in 2
segments, the largest of which is 2015 (1007.5 KiB) in size.
C

```

Figure. Confirming GPT format

2. Use the `dd` utility to dump the Protective MBR and GPT into a file in your home directory. The dump should contain up to first partition entry (Inclusive).

```

root@PXESERVER:~/sfp/efi# ls
gpt_head_entry.bin  share  vboxpostinstall.sh
root@PXESERVER:~/sfp/efi# hexdump -C gpt_head_entry.bin | head
00000000 eb 63 90 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.c.....|
00000010 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
*
00000050 00 00 00 00 00 00 00 00 00 00 00 80 00 08 00 00 |.....|
00000060 00 00 00 00 ff fa 90 90 f6 c2 80 74 05 f6 c2 70 |.....t...p|
00000070 74 02 b2 80 ea 79 7c 00 00 31 c0 8e d8 8e d0 bc |t...yl...1....|
00000080 00 20 fb a0 64 7c 3c ff 74 02 88 c2 52 bb 17 04 |...dl<t...R...|
00000090 f6 07 03 74 06 be 88 7d e8 17 01 be 05 7c b4 41 |...t...l.A|
000000a0 bb aa 55 cd 13 5a 52 72 3d 81 fb 55 aa 75 37 83 |...U.ZR=.U.u7.|
000000b0 e1 01 74 32 31 c0 89 44 04 40 88 44 ff 89 44 02 |.t21..D.e.D..|
root@PXESERVER:~#

```

Figure. Generating GPT dump using `dd`
 I used `cp` and shared folder to download this dump file to my host machine.

3. Load the dump file into a hex dump utility (e.g. 010 editor) to look at the raw data in the file.

```

root@PXESERVER:~# hexdump -C gpt_head_entry.bin
00000000 eb 63 90 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.c.....|
00000010 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
*
00000050 00 00 00 00 00 00 00 00 00 00 00 80 00 08 00 00 |.....|
00000060 00 00 00 00 ff fa 90 90 f6 c2 80 74 05 f6 c2 70 |.....t...p|
00000070 74 02 b2 80 ea 79 7c 00 00 31 c0 8e d8 8e d0 bc |t...yl...1....|
00000080 00 20 fb a0 64 7c 3c ff 74 02 88 c2 52 bb 17 04 |...dl<t...R...|
00000090 f6 07 03 74 06 be 88 7d e8 17 01 be 05 7c b4 41 |...t...l.A|
000000a0 bb aa 55 cd 13 5a 52 72 3d 81 fb 55 aa 75 37 83 |...U.ZR=.U.u7.|
000000b0 e1 01 74 32 31 c0 89 44 04 40 88 44 ff 89 44 02 |.t21..D.e.D..|
000000c0 c7 04 10 00 66 8b 1e 5c 7c 66 89 5c 08 66 8b 1e |...f.Nlf.N.f..|
000000d0 60 7c 66 89 5c 0c c7 44 06 00 70 b4 42 cd 13 72 |lf.N..D..p.B..|
000000e0 05 bb 00 70 eb 76 b4 08 cd 13 73 0d 5a 84 d2 0f |...p.v...s.Z..|
000000f0 83 d0 00 be 93 7d e9 82 00 66 0f b6 c6 88 64 ff |...f...d..|
00000100 40 66 89 44 04 0f b6 d1 c1 e2 02 88 e8 88 f4 40 |0ef.D.....@|
00000110 89 44 00 0f b6 c2 c0 e8 02 66 89 04 66 a1 60 7c |.D.....f..f..|
00000120 66 09 c0 75 4e 66 a1 5c 7c 66 31 d2 66 f7 34 88 |f..uNF.Nlf.f.4.|
00000130 d1 31 d2 66 f7 74 04 3b 44 08 7d 37 fe c1 88 c5 |.1.f.t.:D.J?..|
00000140 30 c0 c1 e8 02 08 c1 88 d0 5a 88 c6 bb 00 70 8e |0.....Z...p..|
00000150 c3 31 db b8 01 02 cd 13 72 1e 8c c3 60 1e b9 00 |.1.....r.....|
00000160 01 8e db 31 f6 bf 00 80 8e c6 fc f3 a5 1f 61 ff |.1.....a..|
00000170 26 5a 7c be 8e 7d eb 03 be 9d 74 e0 34 00 be a2 |821..).4...|
00000180 7d e0 2e 00 cd 10 eb fe 47 52 55 42 20 00 47 65 |D.....GRUB_Ge|
00000190 6f 6d 00 40 61 72 64 20 44 69 73 6b 00 52 65 61 |om.Hard Disk.Rea|
000001a0 64 00 20 45 72 72 6f 72 0d 0a 00 bb 01 00 b4 0e |ld..Error.....|
000001b0 cd 10 ac 3c 00 75 f4 c3 00 00 00 00 00 00 00 00 |...<u.....|
000001c0 02 00 ee ff ff 01 00 00 00 ff ff 1f 03 00 00 |.....|
000001d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
*
000001f0 00 00 00 00 00 00 00 00 00 00 00 00 00 55 aa |.....U..|
00000200 45 46 49 20 50 41 52 54 00 00 01 00 5c 00 00 00 |EFI PART...N...|
00000210 e6 bc 34 29 00 00 00 00 01 00 00 00 00 00 00 00 |...4).....|
00000220 ff ff 1f 03 00 00 00 00 22 00 00 00 00 00 00 00 |.....".....|
00000230 de ff 1f 03 00 00 00 00 63 44 c1 05 1a d2 d1 44 |.....cD...D|
00000240 93 49 41 38 7f f5 45 05 02 00 00 00 00 00 00 00 |.fa8.....|
00000250 80 00 00 00 80 00 00 00 c3 13 a7 15 00 00 00 00 |.....|
00000260 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
*
00000400 48 61 68 21 49 64 6f 6e 74 4e 65 65 64 45 46 49 |Hah!IdontNeedEFI|
00000410 2e b4 c3 25 41 65 5b 4f 85 ab d9 3a 76 76 88 f3 |...%ae10...u...|
00000420 00 08 00 00 00 00 00 00 ff 0f 00 00 00 00 00 00 |.....|
00000430 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
*

```

Figure. Import GPT dump to hexdump
 4. Understand and fully annotate the Protective MBR, GPT header

- 4.1. At what byte index from the start of the disk do the partition table entries start?

Records begin at LBA 2 at 1024 bytes or 0x400 position.

4.2. At what byte index would the partition table start if your server had a so-called "4K native" (4Kn) disk?

For a 4Kn disk, sectors are 4096 bytes, so LBA2 starts at 8192 bytes or 0x2000 position.

Offset (hex)	Field	Bytes (hex)	Value (LE)	Description
0x000 - 0x1BD	x86 startup code section	(varies)	—	MBR bootstrap / manufacturer message / standard loader area
0x1BE - 0x1CD	Partition Record 0 (MBR entry)	02 00 00 00 ... (full entry follows)	boot flag = 0x02 (decimal 2)	Boot flag 0x02 = non-startable in CHS context (as given)
(inside PartRec0)	start CHS	00 ee ff	—	Traditional CHS start bytes
(inside PartRec0)	partition type (category)	EE	—	0xEE = GPT protective MBR partition
(inside PartRec0)	end CHS	ff ff ff	—	CHS end = fully extended / not meaningful for large disks
(inside PartRec0)	start LBA (4 bytes, LE)	01 00 00 00	1	Partition begins at LBA 1 (protects GPT at LBA 1)
(inside PartRec0)	sectors count (4 bytes, LE)	ff ff ff 04	0x04FFFFFF = 83,886,079	Number of sectors in partition → covers almost entire disk (≈40 GiB example)
0x1CE - 0x1FD	Partition Records 1–3	00 ... 00 (all zero)	—	Unused / empty MBR partition entries
0x1FE - 0x1FF	MBR signature	55 AA	—	MBR magic marker (BIOS/MBR detection)
Primary GPT header (LBA 1, bytes 512–1023)				
0x200 - 0x207	GPT signature	45 46 49 20 50 41 52 54	ASCII "EFI PART"	GPT header marker
0x208 - 0x20B	Version	00 00 01 00	0x00000100 = 1.0	GPT header version 1.0
0x20C - 0x20F	Header length	5C 00 00 00	0x5C = 92	Header size = 92 bytes
0x210 - 0x213	Header CRC32	1C DE E0 C6	—	CRC32 of header (with CRC field zeroed when computed)
0x214 - 0x21B	Reserved	00 ... 00	—	Must be zero
0x21C - 0x223	Current LBA (header LBA)	01 00 00 00 00 00 00 00	1	Primary GPT header located at LBA 1
0x224 - 0x22B	Alternate LBA (backup header)	FF FF FF 04 00 00 00 00	0x00000004FFFFFF = 83,886,079	Backup header located at disk end (same as sectors count)
0x22C - 0x233	First usable LBA	22 00 00 00 00 00 00 00	34	First LBA usable for partitions (LBA 34)

Offset (hex)	Field	Bytes (hex)	Value (LE)	Description
0x234 - 0x23B	Last usable LBA	DE FF FF 04 00 00 00 00	0x00000004FFFFFFDE = 83,886,046	Last usable LBA for partitions
0x23C - 0x24B	Disk GUID	23 A3 11 D8 27 0D 92 4C 99 FE 23 91 49 91 C3 C8	—	Disk unique GUID (16 bytes)
0x24C - 0x253	Partition entries starting LBA	02 00 00 00 00 00 00 00	2	Partition array begins at LBA 2
0x254 - 0x257	Number of partition entries	80 00 00 00	128	Count of partition records in the array
0x258 - 0x25B	Size of each partition entry (bytes)	80 00 00 00	128	Each partition record is 128 bytes
0x25C - 0x25F	Partition array CRC32	7F E4 09 02	—	CRC32 over the whole partition array
0x260 - 0x3FF	Reserved	(zeros / reserved)	—	Space reserved by header
First GPT partition record (LBA 2, bytes 1024–1151)	—	—	—	First 128-byte partition entry
0x400 - 0x40F	Partition type GUID	48 61 68 21 49 64 6F 6E 74 4E 65 65 64 45 46 49	—	Partition type / category GUID (16 bytes)
0x410 - 0x41F	Unique partition GUID	4F A4 29 34 97 8C 8A 43 80 0E 1D 6F 98 4D 52 8B	—	Unique GUID for this partition
0x420 - 0x427	Partition first LBA	00 08 00 00 00 00 00 00	2048	Partition starts at LBA 2048
0x428 - 0x42F	Partition last LBA	FF 0F 00 00 00 00 00 00	4095	Partition ends at LBA 4095
0x430 - 0x437	Attribute flags	00 ... 00	0	Partition attribute bits = 0
0x438 - 0x47F	Partition label (UTF-16LE)	00 ... 00	—	Human-readable label (all zeros = empty)

4.3 Name two differences between primary and logical partitions in an MBR partitioning scheme.

In an MBR partitioning system, up to four primary partitions are allowed, but one can be designated as “extended” to serve as a holder for logical partitions. Logical partitions have no quantity limit. Another key distinction is that OS can only be placed on primary partitions, which are directly startable.

References

1. PXE server : [https://www.manageengine.com/products/os-deployer/pxe-preboot-execution-environment.html#:~:text=A Preboot Execution Environment \(PXE,a%20CD%20or%20hard%20disk.](https://www.manageengine.com/products/os-deployer/pxe-preboot-execution-environment.html#:~:text=A Preboot Execution Environment (PXE,a%20CD%20or%20hard%20disk.)
2. dnsmasq configuration : <https://wiki.archlinux.org/title/Dnsmasq>

3. Diagrams: https://tianocore-docs.github.io/Understanding_UEFI_Secure_Boot_Chain/draft/secure_boot_chain_in_uefi/uefi_secure_boot
4. MBR: https://en.wikipedia.org/wiki/Master_boot_record

▼ GPT prompt to assist

You are my technical assistant for creating and executing university-level Unix/Linux labs.
You must always produce:

1. A concise but complete **lab plan**
2. All required **Bash scripts**, always full working code, never placeholders
3. Exact **commands** step-by-step
4. Clear **expected outputs**
5. **Troubleshooting guidance** for every step
6. Explanations written in a strict, logical, no-nonsense tone

Environment assumptions

- Host: Ubuntu Server 22.04
- Virtualization: VirtualBox
- Guest VMs are clean Ubuntu Server installations
- Networking is **configurable** (NAT/bridged/host-only)
- I provide an input text file with tasks

When I send you a task file:

1. Rewrite the tasks into a clean, structured lab plan
2. Generate all required scripts in full
3. Provide complete setup instructions
4. Highlight mistakes or missing **information** (without guessing)
5. Create a final checklist I can follow during execution

Output formatting rules

- Start with a **TL;DR** summary
- Use detailed step-by-step reasoning
- Put all commands, configs, or scripts inside fenced code blocks like:

```
```bash
Example command block
sudo apt update
```
```

Do NOT:

- Use placeholders
- Produce incomplete scripts
- Skip critical context
- Be vague

Wait for me to provide the input file to begin.

